

Explicit sequences



- 1 a Finite b Infinite
- 2 6
- 3 196
- 4
- | | | | |
|---|---|---|---|
| a | $\frac{5}{4}, \frac{6}{5}, \frac{7}{6}, \frac{8}{7}, \frac{9}{8}$ | b | $\frac{2}{3}, 1, 1, \frac{19}{21}, \frac{4}{5}$ |
| c | $-1, -\frac{1}{4}, 0, \frac{1}{16}, \frac{1}{16}$ | d | 0, 3, 6, 9, 12 |
| e | -2, -4, -8, -16, -32 | f | -6, 24, -72, 192, -480 |
| g | 1, 0.7071, 0.5774, 0.5, 0.4472 | h | 2, 2.25, 2.3704, 2.4414, 2.4883 |
- 5 $u_n = 3 \times 0.1^{n-1}$
-

Applications

- 6
- | | | | |
|---|---------------------|---|------------------------|
| a | \$1, \$3, \$7, \$15 | b | \$38, \$43, \$48, \$53 |
| c | Day 6 | | |
- 7
- | | | | |
|---|----------------|---|----|
| a | $u_n = 2n + 7$ | b | 27 |
|---|----------------|---|----|
- 8
- | | | | |
|---|---------------------------|---|-------------|
| a | $u_n = 8.25 (0.55)^{n-1}$ | b | 0.75 metres |
|---|---------------------------|---|-------------|